

B.Tech. Degree IV Semester Examination April 2014**CS 1406 DATA COMMUNICATIONS**
(2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 x 5 = 40)

- I. (a) What are the advantages of digital transmission?
- (b) Explain the various types of communication system.
- (c) Write a note on block coding.
- (d) Discuss briefly about QAM.
- (e) What is piggybacking?
- (f) Write a note on HDLC.
- (g) How is interference avoided by using FDM?
- (h) What is spread spectrum? List out at least three applications of spread spectrum technique.

**PART B**

(4 x 15 = 60)

- II. (a) Compare the features of digital transmission over analog transmission. (10)
 - (b) Explain how bandwidth and SNR have influence on channel capacity. (5)
- OR**
- III. Explain any three guided transmission media. (15)
- IV. (a) Explain the different steps in PCM, with the help of a neat diagram, (8)
 - (b) Write short notes on: (i) B8ZS (ii) HDB3 (7)
- OR**
- V. Describe different digital signal encoding formats. Represent the digital data 001001110011, with the help of neat diagrams. (15)
- VI. Explain how error detection is done using CRC. Implement above scheme for the message 100100 for the generator polynomial $x^3 + x^2 + 1$ (15)
- OR**
- VII. (a) Explain, with suitable example, how single bit error can be corrected using Hamming code algorithm (8)
 - (b) List and explain the three versions of ARQ. (7)
- VIII. (a) Explain in detail frequency hopping spread spectrum. (10)
 - (b) What is CDMA? (5)
- OR**
- IX. (a) What is multiplexing and de-multiplexing? (5)
 - (b) What is DSL? Describe ADSL by defining its modulation technique. (10)